NoteVault Secure Build Report

Threat model, vulnerability findings, remediation and supply-chain controls

Target: NoteVault, Flask + SQLite

Members
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Team: GLASS
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GitHub repository
github.com/march006/notevault-glass

Demonstration video
[Add unlisted YouTube or link-accessible Drive URL]

AI-tool usage disclosure

Tool: ChatGPT / Codex | **How it was used**: Generated the initial app scaffold, vulnerable baseline, threat model draft, vulnerability mappings, remediation code, CI workflow, SBOM draft, and report draft. Team members must review, run, adapt, and be prepared to explain every change.

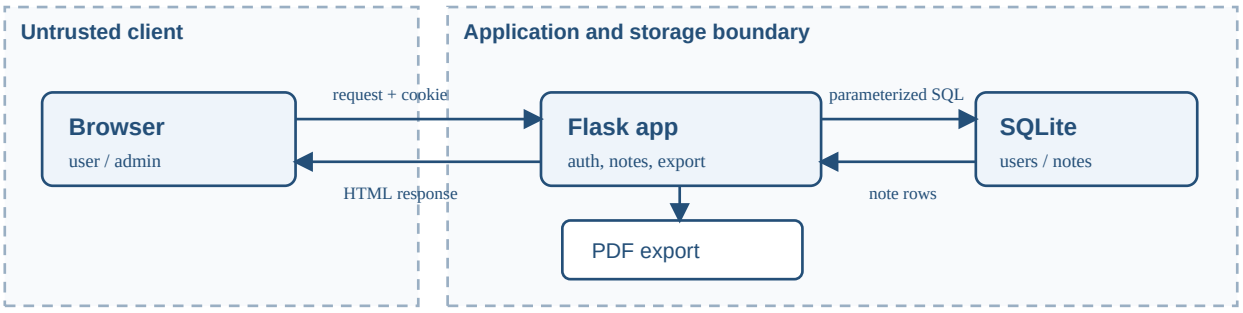
1. Executive Summary

NoteVault is a small Flask note-sharing app. The intentionally vulnerable baseline has eight findings: six High and two Medium. The most serious risks are SQL injection, broken access control, and stored XSS. The remediated build uses parameterized SQL, owner-scoped reads, Jinja escaping, password hashing, CSRF protection, secure session handling, pinned runtime dependencies, and a non-root hardened container. GitHub Actions run #5 passed SAST, SCA, secret scanning, image build, CycloneDX SBOM generation, keyless Cosign signing, and verification of the signed image archive and CycloneDX attestation. Residual risks include no production rate limiter, audit trail, or SLSA provenance predicate.

2. System Scope & Function

The app supports registration and login, creating and viewing personal notes, searching notes, an administrator user list, and PDF export. In scope are browser-to-Flask requests, cookie sessions, SQLite storage, the PDF export function, Python dependencies, Docker image, and GitHub Actions. The vulnerable baseline is isolated under `project/vulnerable-baseline`; the fixed implementation is `project/starter-app`.

Data-Flow Diagram



`TEAM_ID=GLASS` seeds a deterministic traceability marker in the `team_marker` table and displays it in the app. The marker is evidence for this team build, not a credential.

3. Threat Model — STRIDE

Element / data flow: Browser ↔ Flask session and forms | **S**: Credential/session spoofing | **T**: Forged note or login request | **R**: User disputes note change | **I**: Cookie or note disclosure | **D**: Request flooding | **E**: Role escalation | **Top threat & mitigation**: Hash passwords, rotate session, HttpOnly/SameSite/Secure, CSRF tokens, role gate, bounded input

Element / data flow: Flask ↔ SQLite | **S**: N/A | **T**: SQL injection or unauthorized update | **R**: No audit trail | **I**: Cross-user note read | **D**: Expensive query | **E**: Database role abuse | **Top threat & mitigation**: Parameterized SQL, owner_id in every note query, bounded search, add audit trail before production

Element / data flow: Flask → PDF export | **S**: N/A | **T**: Malformed export input | **R**: No export audit | **I**: Another user's note exported | **D**: Large export | **E**: N/A | **Top threat & mitigation**: Owner-scoped note lookup and request-size limits

Element / data flow: Build workflow → artifact | **S**: Compromised action/dependency | **T**: Tampered build or SBOM | **R**: Weak provenance | **I**: Secret leakage | **D**: Resource exhaustion | **E**: Artifact substitution | **Top threat & mitigation**: Scan dependencies, secret scan, short-lived OIDC signing, retain workflow identity and commit SHA

Top 5 risks ranked by likelihood × impact

Rank: 1 | **Risk**: SQL injection in baseline search | **Likelihood**: High | **Impact**: High | **Mitigation**: Parameterized query; SAST/SCA gate

Rank: 2 | **Risk**: IDOR / missing authorization on note and export | **Likelihood**: High | **Impact**: High | **Mitigation**: Owner-scoped access checks and admin decorator

Rank: 3 | **Risk**: Stored XSS in note display | **Likelihood**: Medium | **Impact**: High | **Mitigation**: Template autoescaping; no raw HTML rendering

Rank: 4 | **Risk**: Plaintext credentials and fixed session signing key | **Likelihood**: Medium | **Impact**: High | **Mitigation**: Strong password hashes; required random secret; rotate session

Rank: 5 | **Risk**: Vulnerable dependencies and root container | **Likelihood**: Medium | **Impact**: High | **Mitigation**: Pinned direct dependencies, audit gate, non-root container, SBOM and signing

4. Vulnerability Findings

Severity is qualitative for the local teaching app. Reproduce only against the deliberately vulnerable local baseline with dummy data.

Finding F-01 — SQL injection in search

• **CWE:** CWE-89. **OWASP 2025:** A05 Injection. **Severity:** High.

• **Location:** `project/vulnerable-baseline/app.py`, `/search`.

• **Reproduction:** Start the baseline and request `/search?q=%25%27%20OR%201%3D1%20--%20`. The formatted SQL changes its predicate.

• **Impact:** Search may return records outside the intended query.

• **Evidence:** Source line builds SQL using string formatting; preserve a local request/response screenshot when you record the demo.

Finding F-02 — Broken access control / IDOR

• **CWE:** CWE-639. **OWASP 2025:** A01 Broken Access Control. **Severity:** High.

• **Location:** Baseline `/export/<id>` and `/note/<id>`.

• **Reproduction:** Request `/export/1` without a session, then change the numeric ID to another seeded note.

• **Impact:** Unauthorized note disclosure.

• **Evidence:** Baseline handlers load by id without owner verification; capture the response locally.

Finding F-03 — Stored cross-site scripting

• **CWE:** CWE-79. **OWASP 2025:** A05 Injection. **Severity:** High.

• **Location:** Baseline `/note` and `/note/<id>`.

• **Reproduction:** Sign in to the local baseline, POST title `Demo` and body `<img src=x onerror=alert(1)>` to `/note`, then open the saved note.

• **Impact:** Script executes in a viewer's browser origin.

• **Evidence:** Baseline passes stored title/body into `render_template_string` as HTML; capture browser behavior.

Finding F-04 — Plaintext passwords and known demo admin credential

• **CWE:** CWE-256, CWE-521. **OWASP 2025:** A04 Cryptographic Failures and A07 Authentication Failures. **Severity:** High.

• **Location:** Baseline `users` schema and `/login`; seeded account `admin / admin123`.

• **Reproduction:** Sign in locally using the seeded demo credential or inspect the SQLite `users` table.

• **Impact:** Password disclosure and administrator account takeover.

• **Evidence:** Baseline stores the password field directly and embeds a known development password.

Finding F-05 — CSRF on note creation

• **CWE:** CWE-352. **OWASP 2025:** A01 Broken Access Control. **Severity:** Medium.

• **Location:** Baseline POST `/note`.

• **Reproduction:** While signed in, submit a cross-origin form POST to `/note`; the route has no CSRF token.

• **Impact:** A victim's browser can create an unintended note using the active session.

• **Evidence:** Baseline form and handler contain no CSRF check; record a local request/response.

Finding F-06 — Fixed session signing key and weak cookie policy

• **CWE:** CWE-384, CWE-1275. **OWASP 2025:** A07 Authentication Failures and A02 Security Misconfiguration. **Severity:** Medium.

• **Location:** Baseline Flask initialization and cookie defaults.

• **Reproduction:** Inspect the hard-coded `app.secret_key='dev-secret'` and cookie response flags.

• **Impact:** A known signing secret enables session forgery; weak cookie attributes increase theft/fixation risk.

• **Evidence:** Baseline source shows the fixed key; capture `Set-Cookie` flags for the demo record.

Finding F-07 — End-of-life dependencies

• **CWE:** CWE-1104. **OWASP 2025:** A03 Software Supply Chain Failures. **Severity:** High.

• **Location:** `project/vulnerable-baseline/needs-vulnerable.txt` and Python 3.7 baseline image.

• **Reproduction:** In a disposable environment run `pip-audit -r project/vulnerable-baseline/needs-vulnerable.txt` and inspect the old pinned packages.

• **Impact:** Known vulnerable components can enable compromise.

• **Evidence:** Obsolete versions are explicitly pinned in the baseline; retain scanner output when available.

Finding F-08 — Insecure Docker build configuration

• **CWE:** CWE-250, CWE-1104. **OWASP 2025:** A02 Security Misconfiguration and A03 Software Supply Chain Failures. **Severity:** High.

• **Location:** Baseline Dockerfile.

• **Reproduction:** Inspect the end-of-life Python base, root user, and broad build context.

• **Impact:** Known runtime issues and unnecessary container privileges increase compromise impact.

• **Evidence:** Dockerfile uses Python 3.7 and runs as root.

5. Remediation

Finding: F-01 | **Vulnerable → fixed implementation**: String-formatted SQL → parameterized SQLite query | **Why this fixes it**: User input stays data, not SQL syntax | **Commit**: `fe38f23`

Finding: F-02 | **Vulnerable → fixed implementation**: ID-only lookup → owner-scoped note/export query and admin decorator | **Why this fixes it**: Each request checks the authenticated owner or role | **Commit**: `fe38f23`

Finding: F-03 | **Vulnerable → fixed implementation**: Raw HTML concatenation → Jinja autoescaping | **Why this fixes it**: HTML metacharacters render as text | **Commit**: `fe38f23`

Finding: F-04 | **Vulnerable → fixed implementation**: Plaintext/default credential → Werkzeug password hashes; environment-provided admin password | **Why this fixes it**: Stored values are one-way hashes; no shipped production password | **Commit**: `fe38f23`

Finding: F-05 | **Vulnerable → fixed implementation**: No CSRF token → per-session token on all state-changing forms | **Why this fixes it**: Cross-site requests lack the valid token | **Commit**: `fe38f23`

Finding: F-06 | **Vulnerable → fixed implementation**: Fixed signing key → required secret, session rotation, HttpOnly/SameSite and Secure-on-HTTPS | **Why this fixes it**: Attackers cannot derive signing key from source; cookie handling improves | **Commit**: `fe38f23`

Finding: F-07 | **Vulnerable → fixed implementation**: Old pins → maintained pinned direct dependencies (Flask 3.1.3 and Werkzeug 3.1.9 after the SCA findings), pip-audit gate, CycloneDX SBOM | **Why this fixes it**: Known dependency advisories fail the workflow | **Commit**: `b71fcb6`

Finding: F-08 | **Vulnerable → fixed implementation**: Root/EOL image → patch-version slim image, non-root UID, writable app-owned data directory, read-only FS, dropped capabilities | **Why this fixes it**: Reduces known vulnerable base exposure and runtime privilege while allowing SQLite persistence | **Commit**: `db621ff`, `295d35d`

Reproduction evidence and paired screenshots are included in `project/evidence/attack-vs-fix.md` and the Evidence Appendix below. Requests used loopback-only services and synthetic fixture notes; no real data was used.

6. Supply-Chain Hardening

• **SBOM:** `project/starter-app/sbom.cdx.json` is a CycloneDX 1.6 document listing the four pinned direct runtime requirements: Flask 3.1.3, Werkzeug 3.1.9, ReportLab 4.4.4, and Gunicorn 23.0.0. Workflow run #4 generated the release SBOM from `needs.txt`; the `notevault-sbom` artifact is published with the run.

• **Signing and verification:** Workflow run #5 used GitHub Actions OIDC with Cosign to sign `notevault.tar` and create a CycloneDX attestation for `sbom.cdx.json`. The same step then verified both bundles with the expected identity `https://github.com/march006/notevault-glass/.github/workflows/security.yml@refs/heads/main` and issuer `https://token.actions.githubusercontent.com`; the verification step passed. Rekor transparency-log entries were created (run #4 image blob index 3712304041; run #4 SBOM attestation index 3712304448). The latest bundles are in the `signed-release-material` artifact (about 57 MB); the SBOM is in `notevault-sbom` (about 1.1 KB).

• **SLSA self-assessment:** This build does not claim SLSA conformance yet. CI runs in GitHub Actions and signatures bind artifacts to a short-lived workflow identity, but the workflow does not currently generate a SLSA provenance predicate or meet a hardened builder isolation profile. Add provenance generation and verify builder identity before claiming a SLSA level.

7. Security CI Pipeline

Workflow: `.github/workflows/security.yml`. Bandit runs SAST, pip-audit runs SCA, and Gitleaks scans secrets on push and pull request. Bandit `-lll` fails on High findings; pip-audit fails on reported vulnerabilities; Gitleaks fails on secrets. The supply-chain job builds and signs release material on pushes to `main`. A manual-only `test_high_severity_gate` input appends an ephemeral Bandit HIGH fixture in the runner checkout to demonstrate that the SAST gate blocks it; the fixture is never committed or built.

• **High-severity gate proof:** Workflow run #12 intentionally failed the Bandit SAST step on an ephemeral `subprocess.run(..., shell=True)` fixture (B602, CWE-78, High severity / High confidence). The fixture was added only to the runner checkout by the manual workflow input; it was never committed or built. The supply-chain job was skipped after the scan failure, as intended. The workflow log records the exact finding; the screenshot below captures the failed run summary.

• **Successful run and signed artifact:** Workflow run #11 passed the scans, build, signing, and Cosign verification steps for the current main revision. The earlier run #5 also contains the initial signed artifact and transparency-log evidence.

8. Conclusion & Reflection

The fixed build narrows the main trust-boundary risks with parameterized queries, per-object authorization, output escaping, authenticated state changes, and hardened packaging. Residual production work includes TLS proxy deployment settings, rate limits, audit logging, encrypted backups, dependency lock/verification, and a signed provenance attestation. Team members should document what they learned after reproducing the attacks and reviewing the fixes.

Appendix

• **Tools:** Git, Docker Compose, Bandit, pip-audit, Gitleaks, CycloneDX, and Cosign. The successful run is linked above; exact tool versions are visible in workflow logs.

• **References:** OWASP Top 10 2025; CWE; Sigstore Cosign.

• **Task split:** Thiraphat Kongkan (6631503023) prepared the threat model (DFD, trust boundaries, and STRIDE), tested the vulnerable starter application, and collected attack evidence. Chatmongkhon Jaemjaeng (6631503009) implemented the code remediations and tracked the related commits, hardened the software supply chain and CI (SBOM, Cosign, SAST, SCA, and secret scanning), and compiled the report and demonstration video. Each member submits their evidence-based peer assessment privately as instructed; peer ratings are not included in this public report.

Evidence Appendix

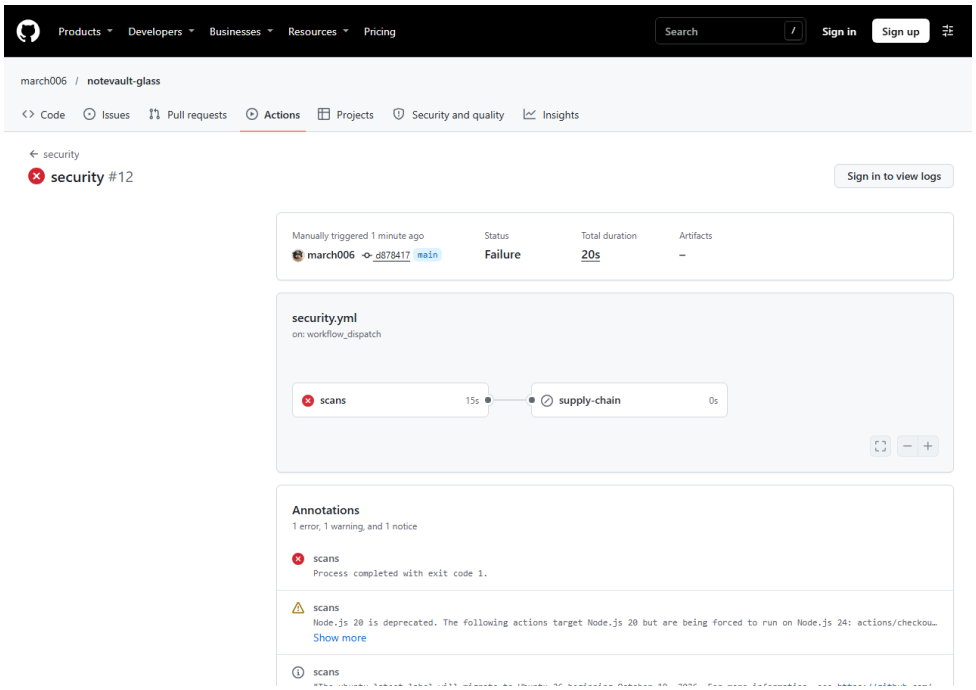
The vulnerable fixture was seeded in a temporary local SQLite database with synthetic records `GLASS-SENTINEL` and `XSS-PROOF` owned by the seeded administrator. Requests were sent to `127.0.0.1:8081`; the remediated Docker app was tested at `localhost:8080`. The secure image now gives `/data` to UID 10001; the registration/login/note flow was re-tested after fixing a volume permission regression.

Finding: F-01 SQL injection | **Observed baseline result**: `q=NOTFOUND` returned no rows; `q=zz' OR 1=1 --` returned both seeded titles. | **Remediated result**: Parameterized, owner-scoped search did not return the other user's sentinel.

Finding: F-02 IDOR / export | **Observed baseline result**: Unauthenticated `GET /export/1` returned HTTP 200 and `PRIVATE-NOTE-PROOF`. | **Remediated result**: Cross-owner export returned HTTP 404.

Finding: F-03 Stored XSS | **Observed baseline result**: `/note/2` rendered the stored `<b>GLASS-XSS-PROOF</b>` as markup. | **Remediated result**: Authenticated secure response escaped it to `<b>GLASS-XSS-PROOF</b>` and displayed it as text.

Finding: F-05 CSRF / role check | **Observed baseline result**: Baseline note form has no CSRF token; the admin route trusts the session role signed with the fixed key. | **Remediated result**: Missing CSRF token returned HTTP 400; a normal user received HTTP 403 from `/admin`.



Detailed requests, responses, and the isolated-fixture note are in `evidence/attack-vs-fix.md`. The Actions screenshot shows the expected failure at the scan job; run #12's Bandit log identifies B602 / CWE-78 at High severity. Run #11 confirms the current revision passes the normal security and release workflow.